Q1. Russell investigated the relationship between mass and weight.
He weighed five different masses using a force meter.

His results are shown in the table.

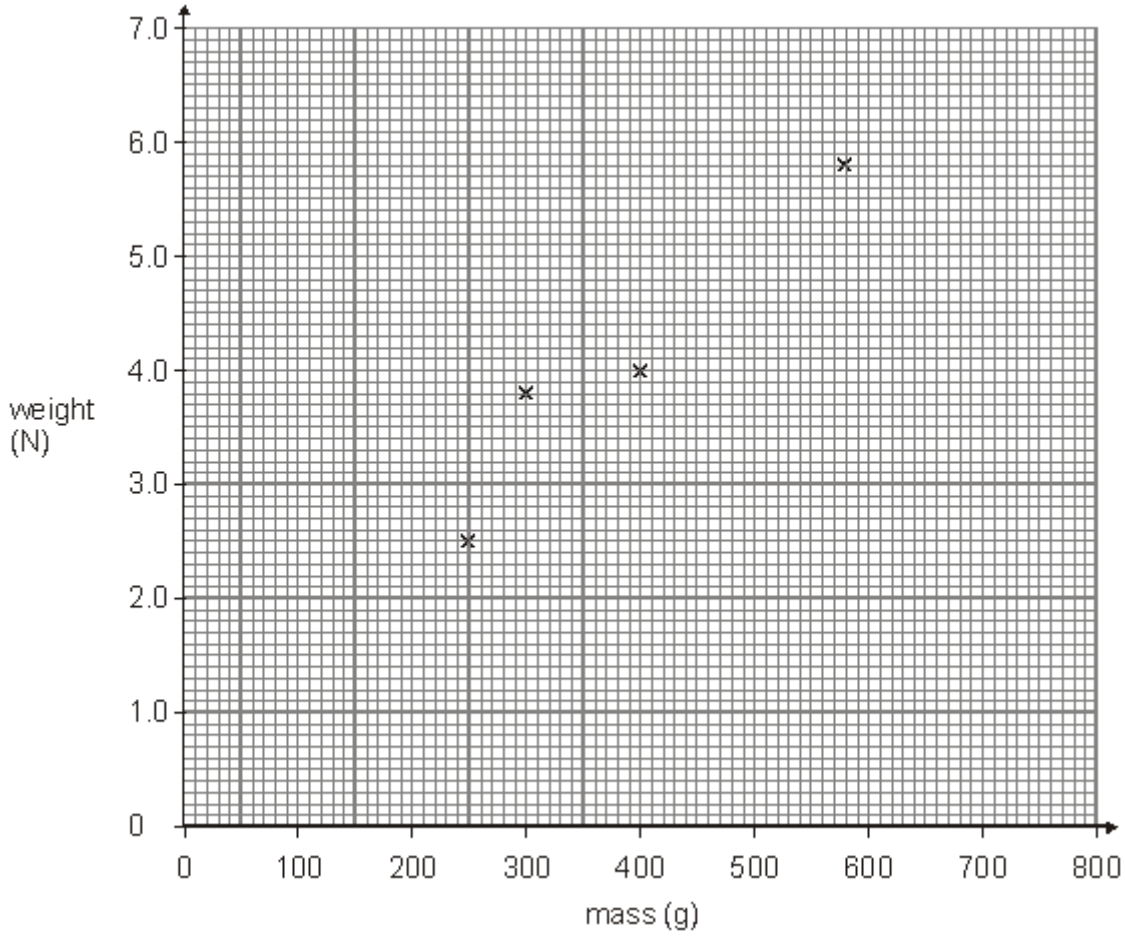
mass (g)	weight (N)
150	1.5
250	2.5
300	3.8
400	4.0
580	5.8

(a) He plotted four of his results on a grid as shown below,

(i) Plot the point for the 150 g mass on the graph.

1 mark

(ii) Draw a line of best fit.



1 mark

(b) One of the points Russell plotted does **not** fit the pattern.

Circle this point on the graph.

1 mark

(c) Use your graph to predict:

(i) the mass of an object weighing 6.5 N;

..... g

1 mark

(ii) the weight of an object of mass 50 g.

..... N

1 mark

(d) Give **one** reason why it is more useful to present the results as a line graph rather than a table.

.....

.....

1 mark

maximum 6 marks

Q1.

(a) (i) point plotted for (150, 1.5) to ± half a small square

1 (L5)

(ii) line of best fit

the anomalous point should be avoided
the line need not be drawn through the origin

1 (L6)

(b) point at (300, 3.8) circled

accept this result circled in the table

1 (L6)

(c) (i) a number from 640 to 660

1 (L6)

(ii) a number from 0.4 to 0.6

*consequential marking applies to both **c i** and **c ii***
accept answers consistent with the graph drawn

1 (L6)

(d) any **one** from

the answer must refer to the results or the pattern shown by the results

- the pattern is revealed **or** observed more easily

  accept 'it allows you to see a pattern'

- it tells you the pattern without working it out

  accept 'you can tell the rule by looking at it'

- it gives readings between the recorded readings

  accept 'it is easier to make predictions'

- you can see if there are results that are wrong **or** do not fit the pattern

  *accept 'it shows better **or** more quickly the more mass the more weight'*

  accept 'the data is continuous'

  *do **not** accept 'it is more accurate **or** precise'*

1 (L6)

[6]